EXPERIMENTAL WORK REPORT

Thin-Film Fabrication and Device Characterization for Neuromorphic Hardware

Summary of five experimental projects: deposition, processing, patterning and electrical characterization

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Research group	Prof. Majid Mohseni
Related thesis	*Multi-Tasking Neuromorphic Processing with Thermal Neuristors* (grade 20/20)
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1. Executive Summary

Five experimental projects were carried out in thin-film fabrication and device measurement, covering every stage from deposition through electrical characterization. The two principal projects target neuromorphic hardware: VO_2 films for thermal neuristors and NiO fiber networks for lithography-free memristive junctions.

Role and ownership. All process steps described in this report were performed by the author. The NiO fiber-network concept is the author's own proposal.

Summary of results

Project	Main outcome	Key figures
1. VO_2 thin films	Clear insulator–metal transition on sapphire	~360× resistance drop (42 kΩ → 118 Ω); T_c ≈ 68 °C (heating), 62 °C (cooling); ~6 K hysteresis
2. NiO fiber networks	Hysteretic, rectifying I–V; fiber morphology controlled by tip-to-collector distance	~5 nA at +10 V; I_+/I_- ≈ 3; ~2 GΩ; NDR-like feature near −7 to −6 V
3. Cu photolithography	Faithful line transfer by subtractive and additive routes	Residual-Cu edge defect identified and traced to contamination / incomplete etching
4. Cu electrodeposition	Conductive Cu coating on graphite	10 V DC, >30 min; matte, granular finish
5. WS_2 precursor-conversion CVD	Two-stage conversion process executed	Films not yet characterized; WS_2 phase unconfirmed

2. Project 1 — VO_2 Thin Films on Sapphire for Thermal Neuristors

2.1 Objective and outcome

Full lab report (PDF) →

Metal sputtering followed by thermal oxidation produced VO_2 films with a clear insulator–metal transition: a ~360× resistance drop (42 kΩ → 118 Ω), T_c ≈ 68 °C on heating and 62 °C on cooling (~6 K hysteresis).

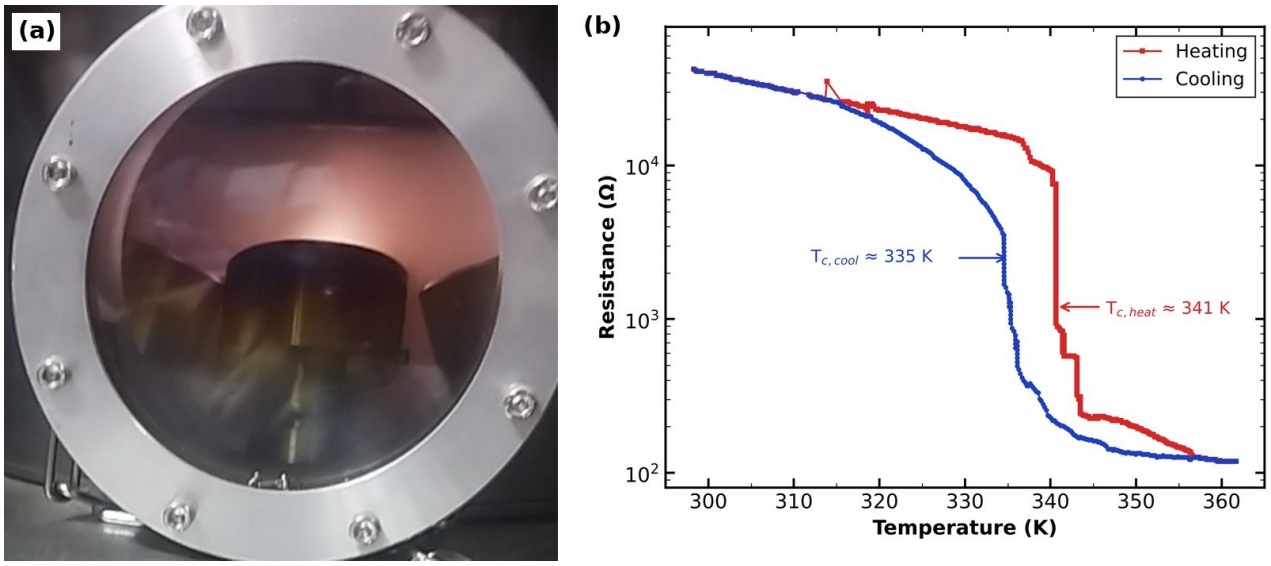


Figure 1. (a) Ar plasma through the viewport of the multi-target RF sputter system used for the V and Cu depositions. (b) Resistance vs. temperature of a VO_2 film on sapphire: ~360× drop and ~6 K hysteresis.

2.2 Method

- **Process:** RF magnetron sputtering of metallic V on c-plane sapphire in pure Ar (no substrate heating, ≥2 h), followed by oxidation in a tube furnace under low-pressure Ar (30 min). This route avoids reactive-gas control entirely.
- **Devices and measurement:** two-terminal devices with Ag-paste contacts (gaps 0.2–3.3 mm); I–V sweeps on a Keithley 2400 probe station; R–T on a heated stage up to ~90 °C.

2.3 Results

- **Electrothermal behaviour:** in a low-resistance device, resistance fell from 50 Ω to 22 Ω as dissipated power rose to 0.25 W (negative-TCR self-heating). Above ~2.4 V the device entered current-controlled NDR, ran away to the 1 A compliance and held a 2–3 Ω ON state until relaxing near 1.0–1.2 V.

2.4 Critical assessment

- R–T and I–V data come from different devices, so the runaway is not yet tied to the insulator–metal transition.
- Room-temperature resistances span six orders of magnitude across devices (possible V_2O_3 or under-oxidized regions, Ag bridging).
- A self-heating fit could not separate activation energy from thermal resistance (correlation −0.985); therefore no activation energy is reported.

2.5 Next steps

- R–T and I–V measurements on the same device; Raman/XRD for phase identification.
- Lithographically defined µm gaps with 4-wire contacts.
- A series resistor with ≤100 mA compliance for repeatable, non-destructive switching.

Relevance. This work connects directly to the thesis, in which coupled VO_2 neuristors were modelled for reservoir computing. A code-based virtual lab simulating this sputtering-to-device workflow was also built.

3. Project 2 — NiO Electrospun Fiber Networks for Lithography-Free Memristive Junctions

3.1 Objective and concept

Full lab report (PDF) →

Own concept: build many-junction memristor arrays without high-resolution lithography, which the laboratory lacked. A NiO fiber layer and an insulating PVP fiber layer are stacked so that every crossing acts as one junction.

3.2 Method

- **Process:** $Ni(NO_3)_2$/PVP aqueous solution; horizontal electrospinning onto Al foil; PVP spacer layer; anneal in a tube furnace under Ar/N_2; transfer to glass; Ag-paste contacts.
- **Parameter study:** six conditions (10–17 cm, 5–15 kV) mapped by optical microscopy and needle-tip video.

3.3 Results

- **Morphology:** with a water-based solution, tip-to-collector distance mattered more than field strength. At the same 1.0 kV/cm, 10 cm gave wet, coiled mats and 15 cm gave thin, straight, bead-free fibers. This was traced to flight time versus slow water evaporation; the absent Taylor cone and intermittent jetting were linked to water's high surface tension and feed-rate imbalance.

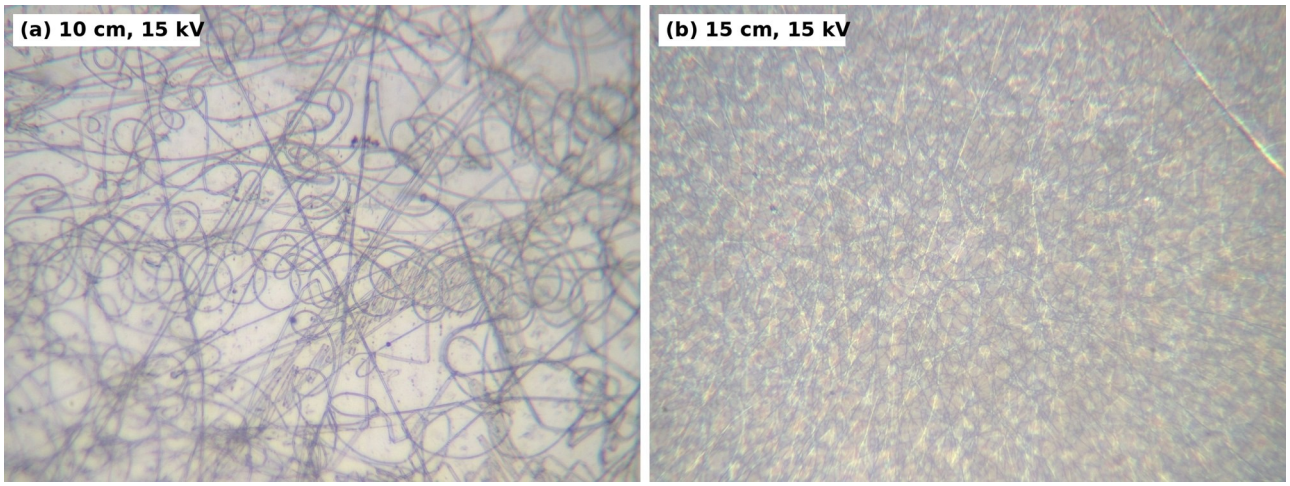


Figure 2. Same voltage, different distance (50×): (a) wet, coiled fibers at 10 cm; (b) dense, straight, bead-free fibers at 15 cm.

- **Device result:** the network showed hysteretic, rectifying I–V over ±10 V at the nA level (~5 nA at +10 V, I_+/I_- ≈ 3, ~2 GΩ), with an NDR-like feature near −7 to −6 V.

3.4 Critical assessment

- Addressable junctions were not achieved: the inert-gas anneal carbonized the PVP spacer, and the dense mat formed a percolating network.
- Likely origins of the hysteresis were ranked, with Ag migration from the paste most likely and intrinsic NiO switching least likely. Separating them requires control samples, XRD and SEM.

3.5 Redesign

- Ethanol/water co-solvent and aligned collection (rotating drum or parallel gap).
- Calcination in air before adding a heat-stable oxide spacer; evaporated electrodes.

Relevance. Random self-assembled memristive networks are studied as physical reservoirs. The nonlinearity, hysteresis and many coupled nodes observed here are the properties a reservoir needs, linking this work to the thesis on reservoir computing.

4. Projects 3–5 — Patterning, Deposition and Thermal Processing

Three shorter projects provided hands-on experience with the fabrication steps on which device work depends.

4.1 Cu photolithography — two routes

Full lab report (PDF) →

Cu films were deposited on glass by RF sputtering and patterned by two routes:

- **Subtractive:** S1813 positive resist + $FeCl_3$ wet etch.
- **Additive:** AZ nLOF 2020 negative resist + lift-off. Because cross-linked nLOF resists acetone, hot NaOH was used for lift-off.

Steps covered spin coating, soft and post-exposure bakes, mask alignment and contact UV exposure under yellow light. Optical microscopy confirmed faithful line transfer and revealed a residual-Cu edge defect that could short adjacent electrodes, traced to contamination and incomplete etching.

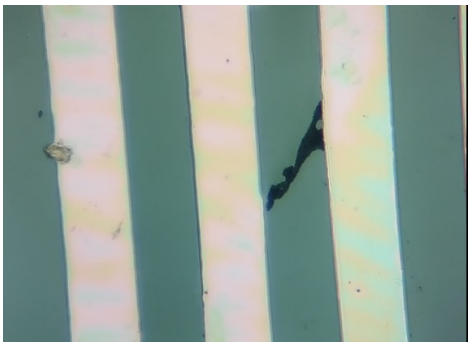


Figure 3. Etched Cu lines (light) on glass: straight edges, a surface particle, and a residual-Cu defect bridging into the gap.

4.2 Cu electrodeposition on graphite

Full lab report (PDF) →

Within a postdoc-led device project, the $CuSO_4$/SDS electrolyte was designed and prepared from stoichiometric calculations, a two-electrode cell was assembled and the deposition was run (10 V DC, >30 min). The coating was conductive. Its matte, granular finish was interpreted as transport-limited growth. Next steps: a three-electrode potentiostatic setup and Faraday-law thickness tracking.

4.3 WS_2 by precursor-conversion CVD

Full lab report (PDF) →

A two-step route was run from dip-coated $(NH_4)_2WS_4$ on Si/SiO_2: vacuum evacuation, Ar purge, Ar/H_2 flow via mass flow controllers, and a programmed two-stage ramp-and-hold profile (decomposition to WS_3, then reduction and crystallization). The films are uncharacterized, so the WS_2 phase is not confirmed.

5. Technical Skills and Equipment

Area	Techniques	Equipment used
Thin-film deposition	RF magnetron sputtering (V, Cu), electrospinning, dip coating, electrodeposition	Six-target RF sputter system (pump-down to 10^{-4} mbar), electrospinning setup, programmable DC supply
Thermal processing	Oxidation and inert-gas anneals, two-stage precursor conversion	Tube furnaces (quartz tube, rotary-pump vacuum), multi-segment controller, mass flow controllers (Ar, H_2)
Lithography	Positive and negative UV resists, spin coating, bakes, contact exposure, wet etching, lift-off	Spin coater, mask aligner with alignment camera, hotplates
Electrical characterization	I–V sweeps, R–T across a phase transition, self-heating and NDR analysis	Keithley 2400 SMU, probe station, heated stage
Inspection and analysis	Optical microscopy, defect analysis, Python data analysis and model fitting	Optical microscopes (20×/50×)